WORKED SOLUTION

STEP III 2006, Question 4

Pure • Functional Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q4

The key idea

The opening identity is Cauchy's functional equation. Putting $y = x$ gives a relation between f at x and at $2x$, and differentiating that relation n times and setting $x = 0$ kills almost every Maclaurin coefficient. Each later part is the same equation in disguise: a change of variable (logarithm, exponential, tangent) turns the new identity back into $(*)$, so we can use the answer we already have.

Solution

The basic equation

Put $y = x$ in $(*)$: $\mathrm{f}(x) + \mathrm{f}(x) = \mathrm{f}(2x)$, that is,

$$2\mathrm{f}(x) \equiv \mathrm{f}(2x).$$

This is an identity in x, so we may differentiate both sides. By the chain rule, each differentiation of $\mathrm{f}(2x)$ brings out a factor of 2, so after n differentiations

$$2\mathrm{f}^{(n)}(x) \equiv 2^n \mathrm{f}^{(n)}(2x).$$

Setting $x = 0$ gives $(2 - 2^n)\,\mathrm{f}^{(n)}(0) = 0$. With $n = 2$ this reads $-2\mathrm{f}''(0) = 0$, so $\mathrm{f}''(0) = 0$, as required.

The same argument works for every $n \neq 1$: the factor $2 - 2^n$ is zero only when $n = 1$, so

$$\mathrm{f}(0) = 0 \quad \text{and} \quad \mathrm{f}^{(n)}(0) = 0 \ \text{ for all } n \geq 2.$$

(The case $n = 0$ is just $2\mathrm{f}(0) = \mathrm{f}(0)$.) Only the linear term of the Maclaurin series survives:

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)\,x + \tfrac{1}{2!}\mathrm{f}''(0)\,x^2 + \cdots = kx, \qquad k = \mathrm{f}'(0).$$

Conversely $kx + ky = k(x + y)$, so every such function works. The most general solution of $(*)$ is $\mathrm{f}(x) = kx$, where k is any constant.

Part (i)

To take a logarithm we need $g > 0$, so first we deal with the sign. Putting $x = y = \frac{1}{2}s$ gives $g(s) = (g(\frac{1}{2}s))^2 \geq 0$. If $g(a) = 0$ for some a, then for every x,

$$g(x) = g(a)\, g(x-a) = 0,$$

so $g \equiv 0$, which does satisfy the identity. Otherwise $g(x) > 0$ for all x, and $G = \ln g$ is defined everywhere. Taking logarithms of $g(x)g(y) = g(x+y)$ gives

$$G(x) + G(y) = G(x+y),$$

which is exactly $(*)$. So $G(x) = kx$, and

$$g(x) = e^{kx}$$

for a constant k, which clearly works since $e^{kx}e^{ky} = e^{k(x+y)}$. Apart from the trivial solution $g \equiv 0$, these are all the solutions.

Part (ii)

Every positive x is e^u for exactly one real u, namely $u = \ln x$. So we can write $x = e^u$ and $y = e^v$, where u and v range over *all* real numbers. Then $xy = e^{u+v}$, and the identity becomes

$$H(u) + H(v) = h(e^u) + h(e^v) = h(e^{u+v}) = H(u+v)$$

for all real u, v. This is $(*)$ again, so $H(u) = ku$. Since $u = \ln x$,

$$h(x) = k \ln x \qquad (x > 0),$$

and this works because $\ln x + \ln y = \ln(xy)$.

Part (iii)

The expression $\dfrac{x+y}{1-xy}$ is the tangent addition formula in disguise. Write $x = \tan u$ and $y = \tan v$ with $u, v \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so that $u = \arctan x$ and $v = \arctan y$. Then

$$z = \frac{\tan u + \tan v}{1 - \tan u \tan v} = \tan(u+v).$$

Define $T(u) = t(\tan u)$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Whenever $u+v$ also lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, the identity says

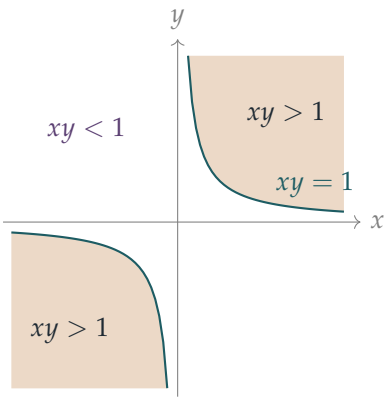
$$T(u) + T(v) = t(x) + t(y) = t(\tan(u+v)) = T(u+v).$$

Ask yourself The argument for $(*)$ used the identity for *all* x and y. Does it still work when we only know it for u, v and $u+v$ in $(-\frac{\pi}{2}, \frac{\pi}{2})$?

Yes. The Maclaurin argument only looks near 0. For $|u| < \frac{\pi}{4}$ we may put $v = u$ (then $|2u| < \frac{\pi}{2}$), which gives $2\mathrm{T}(u) = \mathrm{T}(2u)$ near 0. Differentiating n times at $u = 0$ gives $(2 - 2^n)\mathrm{T}^{(n)}(0) = 0$ exactly as before, so $\mathrm{T}(u) = ku$. Since $u = \arctan x$,

$$\mathrm{t}(x) = k \arctan x.$$

For which x and y does this really work? If $xy < 1$ then $u + v \in (-\frac{\pi}{2}, \frac{\pi}{2})$. When u and v have opposite signs this is automatic. When both are positive, $\tan u \tan v < 1$ means $\tan v < \cot u = \tan(\frac{\pi}{2} - u)$, so $u + v < \frac{\pi}{2}$, and the negative case is the same by symmetry. So $\arctan x + \arctan y = \arctan z$, and the identity holds. If $xy > 1$, however, $u + v$ lies outside $(-\frac{\pi}{2}, \frac{\pi}{2})$, and then $\arctan z = u + v \mp \pi$. For example, $x = y = 2$ gives $z = -\frac{4}{3}$, and $2\arctan 2 - \arctan(-\frac{4}{3}) = \pi$.



So $\mathrm{t}(x) = k \arctan x$ satisfies the identity for every pair with $xy < 1$, the region containing the origin in the figure. That is the answer the question intends. In the shaded regions $xy > 1$, the two sides differ by $\pm k\pi$. If we insist on the identity for literally every pair with $xy \neq 1$, only $k = 0$ survives, and the only solution is $\mathrm{t} \equiv 0$.

Checks

- For part (iii), differentiate the identity with respect to y and then put $y = 0$. Since $\partial z/\partial y = (1 + x^2)/(1 - xy)^2$, which is $1 + x^2$ at $y = 0$, we get $\mathrm{t}'(0) = \mathrm{t}'(x)(1 + x^2)$. So $\mathrm{t}'(x) = k/(1 + x^2)$, and with $\mathrm{t}(0) = 0$ this integrates to $k \arctan x$. The same trick on $(*)$ gives $\mathrm{f}'(x) = \mathrm{f}'(0)$ at once.
- We tested all four answers numerically on 10^4 random pairs each. In part (iii), $k \arctan x$ satisfied the identity on every one of about 5×10^4 random pairs with $xy < 1$, and missed by exactly $k\pi$ in size on every pair with $xy > 1$.

Common slips

- Forgetting the constant term: $\mathrm{f}(0) = 0$ comes from the case $n = 0$, so the answer is kx, not $kx + c$.

- Taking $\ln g$ without checking that $g > 0$. The zero function is a genuine solution of part (i).
- Assuming $\arctan x + \arctan y = \arctan \dfrac{x+y}{1-xy}$ always holds. It needs $xy < 1$.

Beyond STEP

Without the assumption of a Maclaurin series, Cauchy's equation $(*)$ needs much less: if f is merely continuous (or even just bounded on some interval), then $f(x) = kx$ still follows, starting from $f(q) = q\,f(1)$ for rational q. Using the axiom of choice, though, one can build wild solutions that are not linear, whose graphs are dense in the plane. Part (i) also shows that the exponential function is the essentially unique continuous homomorphism from addition to multiplication, an idea that reappears throughout group theory. Abbott's *Understanding Analysis* is a good place to see continuity and density arguments of this kind done carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.